

Design Verification Plan & Report — finN-2e19

Document: VBF-T6R1F-DVPR-01 | **Case:** t6_r1_flash | **Date:** 2026-08-25 | **Model fidelity:** SPM_e (screening) + DFN (final endorsement, task 5)

1. Verification Matrix

#	Criterion	Requirement	Method	Protocol	Result	Verdict
M1	Volumetric energy density	≥ 950 Wh/L	calc-energy (contract formula)	1C_discharge, DFN	1171.7 Wh/L	PASS
M2	Voltage plateau	≥ 4.1 V	midpoint voltage	1C_discharge, DFN	4.145 V	PASS
M3	SEI after 100 cycles	≤ 500 nm	SEI thickness (Yang2017 EC-limited)	aging_1C_100cyc, SPM _e (DFN 1-cyc x-check)	292.2 nm	PASS
M4	4C charge, no plating	plated = false	anode potential min ≥ 0	4C_charge_45C (18 A to 4.7 V, plating model)	−0.019 V (DFN) → plated	FAIL (boundary, see §3)
M5	Max temperature	≤ 50 °C (323.15 K)	T_max (lumped thermal, DFN heat)	4C_charge_45C, 45 °C amb, h=300	322.50 K	PASS

2. Evidence Files (case-relative)

Metric	File	Key
M1, M2	cell/r5_finN_dfn_1c_calc.json	energy_density_wh_l, midpoint_voltage_v
M3	cell/r5_finN2e19_aging.json	sei_thickness_nm_end
M4, M5	cell/r5_finN_dfn_h300_4c.json	anode_potential_v (min), T_max_K
M3 x-check	cell/r5_finN_dfn_aging.json	sei_thickness_nm_end (1 cyc)

Audit trail: log.jsonl entries — propose R1–R5, evaluate R1(2)/R2(5)/R3(5)/R4(4)/R5(10), plan (initial + 2 updates), design finN-2e19, verify DFN.

3. M4 Failure Analysis (documented boundary)

- **Observation:** every configuration with a real 4C charge (5.3–7.4 Ah) shows anode minimum −0.21...−0.26 V (SPM) / −0.02 V (DFN) at the cutoff; every mechanically-passing configuration delivers ≤ 0.003 Ah (degenerate charge, rejected).
- **Root cause:** LNMO OCP ceiling 4.706 V ≡ charge cutoff 4.7 V → at the cutoff the anode potential is pinned by the cathode side ($\phi_{ap} = U_{pos} + \eta_{pos} - 4.7 - \phi_e(\text{sep})$); $\phi_e(\text{sep}) \approx +0.2$ V is dominated by electrolyte potential in the anode (measured 11× ohmic estimate) and cannot be removed with physical parameters ($\sigma \blacksquare 100$ S/m required).
- **Rounds to establish:** 3 consecutive rounds (R3–R5), same cause → three-strike questioning recorded in plan update 2.
- **Decision:** retain a real 6.68 Ah 4C charge (the functional intent of "4C fast charge") and record M4 as fail; degenerate "passes" explicitly not claimed. This is a system/protocol interaction, not a fixable component defect (see DFMEA FM-1).

- **Residual options** (outside this case's degrees of freedom): lower charge cutoff (usage-mode parameter, excluded), different cathode system (electrode system locked by task).

4. DFN Cross-Validation (task 5)

Quantity	SPMe	DFN	Δ
ED (Wh/L)	1172.1	1171.7	−0.03%
Midpoint (V)	4.248	4.145	−2.4%
Capacity (Ah)	6.347	6.324	−0.4%
4C charge (Ah)	6.97	6.68	−4.2%
Anode min (V)	−0.262	−0.019	—
T_max @h120 (K)	321.9	326.1	+4.3 K

SPMe is adequate for ED/capacity screening; **thermal and plating verdicts taken at DFN fidelity.**

5. Pass/Fail Summary

3/4 criteria PASS with margin; M4 FAIL (documented protocol-level boundary). Case verdict: partial pass with honest M4 record.